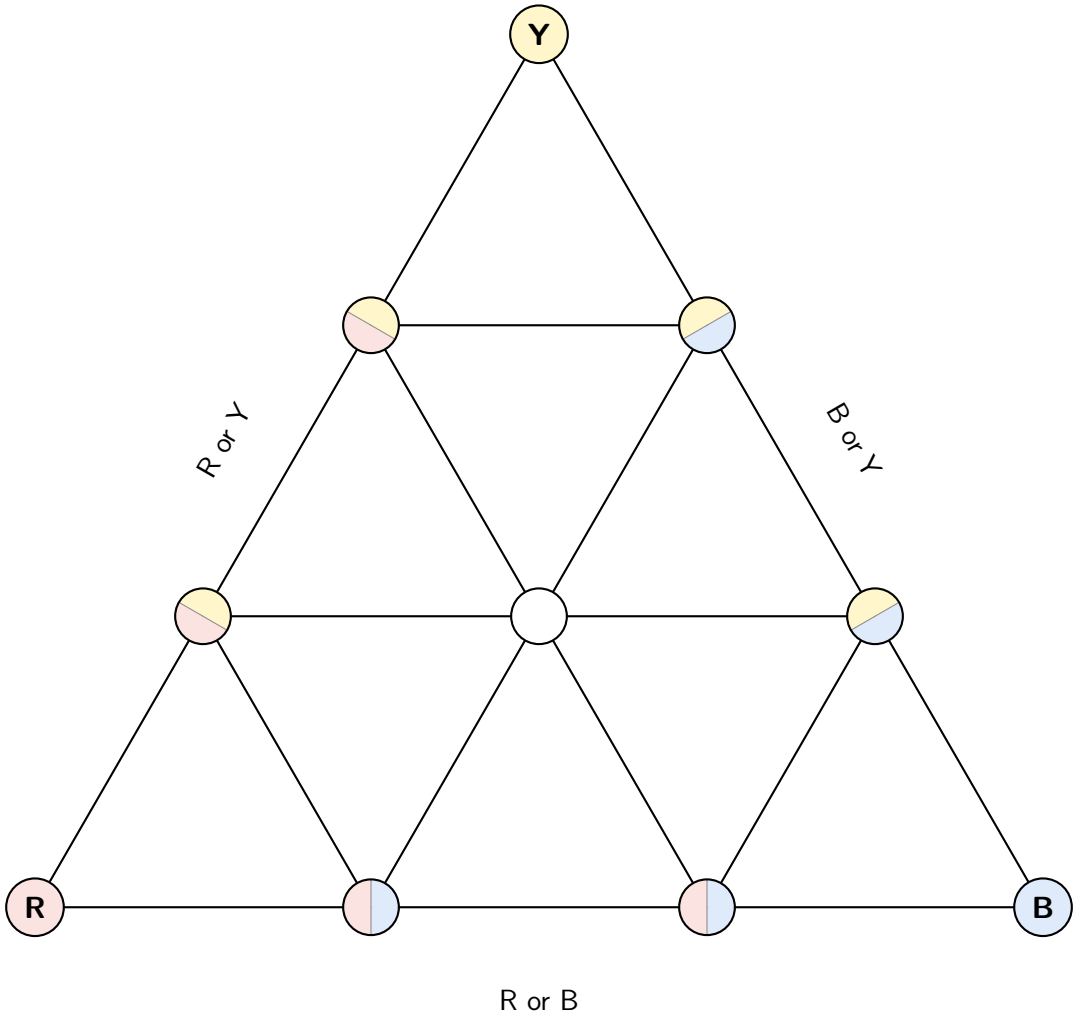
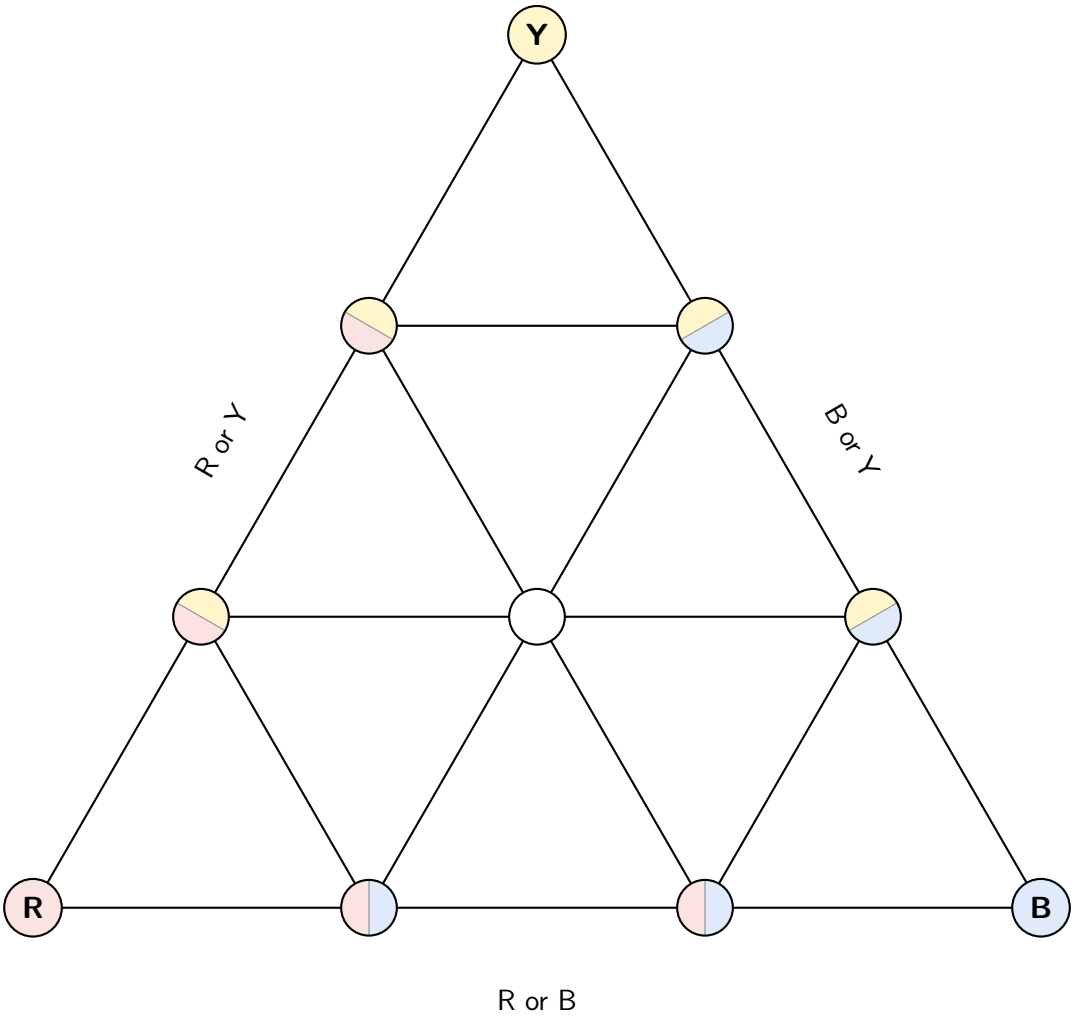
Put R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

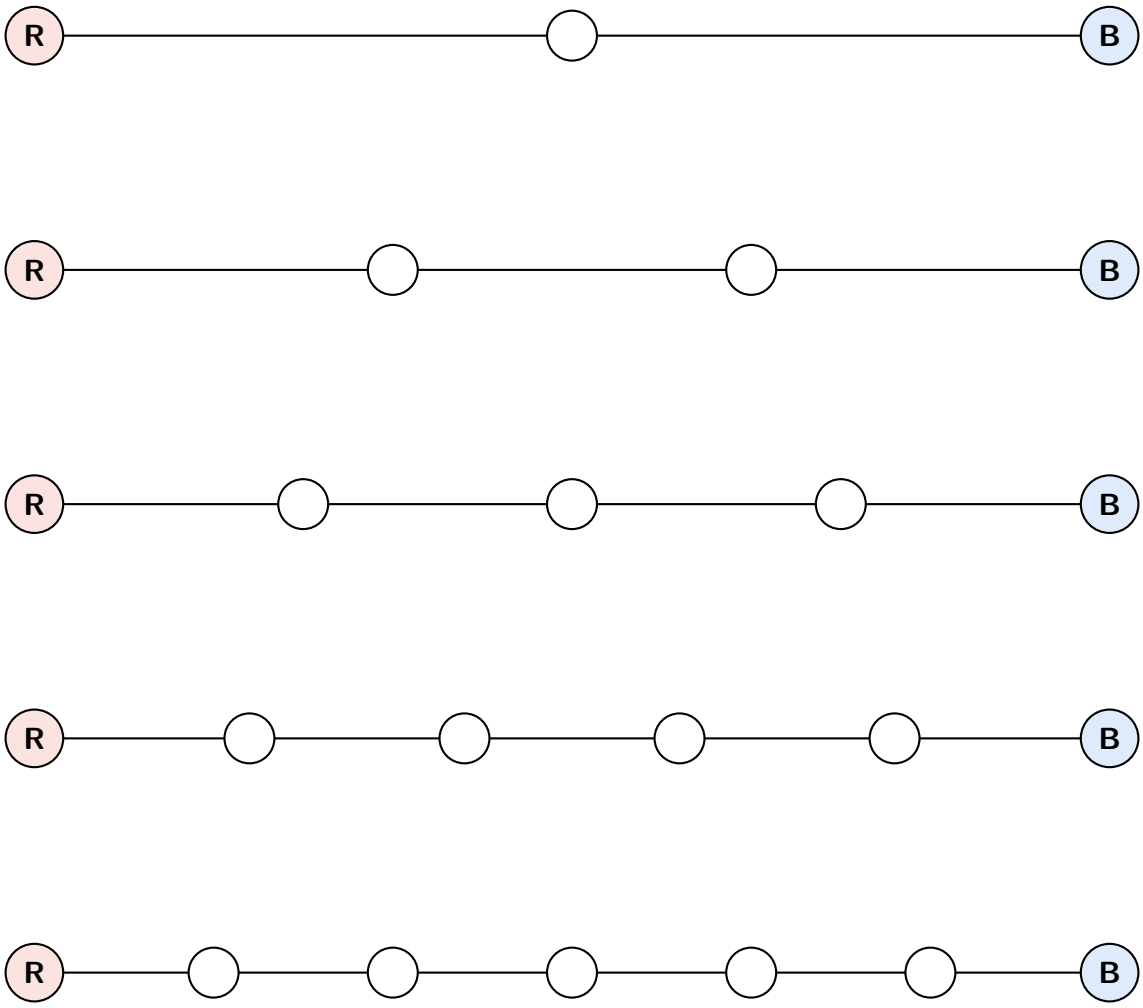
Problem 1: Try fillings with R, B, and Y counters. Find a filling with as few little triangles containing R, B, and Y as you can, and another with as many as you can. Write each record on its own board.



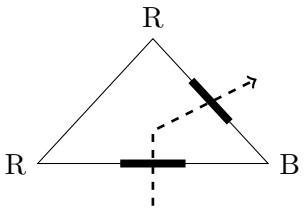
Another board for Problem 1.



Problem 2: An edge with R at one end and B at the other is a door. Write R or B on every empty dot and mark the doors. Can any row have an even number of doors? Explain why or why not.



Problem 4: Mark every door. Draw every route that starts at an outside door. A route must leave a two-door triangle through its other door; it stops at a one-door triangle or outside. Which outside doors lead back outside?



Example: enter one thick door and leave through the other.
The dashed route crosses edges; it does not follow them.

